

The Three-Level Posterior Tower: Hierarchical Bayesian Inference and Mixed Distributive Laws

Revised version

Vinay K. Awasthi
Department of Applied Mathematics and Statistics
Johns Hopkins University

October 3, 2026

Abstract

We construct a categorical framework for structured Bayesian inference that keeps three levels of posterior information as state. The central object is a context comonad $C_3(X) = X \times E_3(X)$, where $E_3(X)$ collects the outcome-, hypothesis- and mixture-level posteriors and a shadow over structured hypotheses. We prove that $X \times E(X)$ is a comonad for *every* endofunctor E of **Set**, and we obtain the context lift and the distribution-level lift as instances of functoriality and naturality, so that their coherence conditions hold by proof rather than by stipulation. We then study mixed distributive laws between the finite distribution monad D and C_3 . The hierarchy-preserving *shift law* λ^σ is natural and satisfies the counit axiom but fails the unit, multiplication and comultiplication axioms; its unit and multiplication defects vanish after barycentric collapse. The *averaging law* $\bar{\lambda}$ satisfies naturality and the unit, counit and multiplication axioms exactly, and fails comultiplication. We show that the unit axiom alone forces level-one hierarchy to be expressed through $D(\eta_X)$ rather than through a point mass at the posterior, which locates the tension between hierarchy and collapse precisely. Whether a mixed distributive law satisfying all four axioms exists for C_3 remains open. On the algebraic side we define a normalized partial composition on structured posteriors, correct the conditioning laws, characterize exactly when observational equivalence is a congruence, and replace a composite-identifiability axiom (which forces the hypothesis space to be trivial) and show that the likelihood-factorization “iff” holds under nondegeneracy alone. All counterexamples and the exact identities were also checked by computation in exact rational arithmetic. The framework is illustrated on the CausalIDView gradient-free causal identification task.

Contents

1	Introduction	2
2	Preliminaries	3
2.1	The finite distribution monad	3
2.2	Mixed distributive laws	3
3	Structured hypotheses	4
4	The structured posterior	4
4.1	Mixture	4
4.2	Composition	5

4.3	Conditioning	5
5	Observational equivalence	6
6	The three-level tower and the context comonad	7
6.1	The three posterior levels	7
6.2	A general context comonad	7
6.3	The three-level context	7
6.4	The context lift	8
7	The distribution-level lift	8
8	Mixed distributive laws for the tower	8
8.1	What the unit axiom forces	8
8.2	The shift law	9
8.3	The averaging law	9
9	Nondegeneracy and the likelihood-factorization criterion	11
10	Empirical setup	11
11	Related work	12
12	Summary of results	12
13	Open problems	12
14	Conclusion	13
A	Empirical tables	13
A.1	The 40-world benchmark	13
A.2	Chain-pooling curve	13
A.3	Coverage	13
A.4	Shallow-MLP worlds	13
A.5	Spectral pre-pass	14
B	Changes from the previous version	14
C	Computational checks	15

1 Introduction

Bayesian inference in structured problems produces a hierarchy of posteriors: a posterior over outcomes, a posterior over hypotheses (each of which itself predicts outcomes), and a posterior over the ways hypotheses combine. This paper builds a categorical framework that retains all three levels as state rather than as intermediate computations, and asks how such state interacts with the probability monad.

The framework builds on the three-layer architecture of [6]; the hierarchy layer is developed in [4] and the holonomy obstruction in [5].

Contributions.

1. A general *context comonad* $C_E(X) = X \times E(X)$ for any endofunctor E of **Set** (theorem 6.1), specialized to the three-level tower C_3 .
2. A hypothesis functor (Hyp, q) with q natural, so that the compatibility square is naturality (section 3); the context lift $J = \text{Hyp}(j)$ and the distribution-level lift K satisfy all coherence conditions by proof (propositions 6.6 and 7.2).
3. A normalized partial algebra $(\oplus, \otimes, \text{Cond})$ on structured posteriors with corrected laws (section 4).
4. An affine first isomorphism theorem for observational equivalence, and an exact characterization of when it is a congruence for $\otimes$ and for Cond (section 5).
5. Two mixed distributive laws λ^σ and $\bar{\lambda}$ with a complete account of which Beck axioms each satisfies (section 8, summarized in table 1), together with a structural lemma showing that the unit axiom forbids point-mass hierarchy at level one (lemma 8.1).
6. A proof that composite identifiability trivializes the hypothesis space, and a proof that the likelihood-factorization “iff” holds under nondegeneracy alone (remark 9.2 and theorem 9.3).

Relation to the previous version. An earlier version of this paper claimed that a three-level transformation $\lambda^{[3]}$ satisfies the unit, counit and comultiplication axioms exactly and extends to a weak distributive law with a barycentric 2-cell. Those claims do not hold; the corrected statements are theorems 8.3, 8.6 and 8.7. A full list of changes is in appendix B.

2 Preliminaries

2.1 The finite distribution monad

Let D be the finitely supported distribution monad on **Set**: $D(X)$ is the set of finitely supported probability measures on X , $D(f) = f_*$ is pushforward, $\eta_X(x) = \delta_x$, and $\mu_X: D^2X \rightarrow DX$ is barycenter, $\mu_X(\sum_i a_i \delta_{\rho_i}) = \sum_i a_i \rho_i$. Its Kleisli category restricted to finite sets is the category of finite stochastic matrices; see [12, 14] for the Markov-category perspective.

We work on **Set** rather than on finite sets: even for finite X with $|X| \geq 2$, the set $D(X)$ is an infinite simplex, so D does not restrict to an endofunctor of finite sets. Finiteness enters only through finite supports.

Each $D(Y)$ is a convex set, and every $D(f)$ is affine. A product of convex sets is convex componentwise, and a product of affine maps is affine.

2.2 Mixed distributive laws

Definition 2.1 (Mixed distributive law [9, 19]). Let (T, η, μ) be a monad and (C, ε, δ) a comonad on the same category. A *mixed distributive law* is a natural transformation $\lambda: TC \Rightarrow CT$ such that

$$\begin{array}{ll} \text{(U)} & \lambda \circ \eta_C = C\eta, \\ \text{(Co)} & \varepsilon_T \circ \lambda = T\varepsilon, \\ \text{(M)} & \lambda \circ \mu_C = C\mu \circ \lambda_T \circ T\lambda, \\ \text{(Cm)} & \delta_T \circ \lambda = C\lambda \circ \lambda_C \circ T\delta. \end{array}$$

Weak distributive laws between two monads, in the sense of Street and Böhm [21, 10, 11], keep the multiplication-type axioms and relax the unit-type ones; Garner [15] and Goy–Petrişan [16] give probabilistic and topological examples. We return to this comparison in remark 8.9.

3 Structured hypotheses

Definition 3.1 (Hypothesis functor). A *hypothesis functor* is a functor $\text{Hyp}: \mathbf{Set} \rightarrow \mathbf{Set}$ with a natural transformation $q: \text{Hyp} \Rightarrow \mathbf{D}$, the *predictive map*. Naturality of q is the compatibility square

$$\begin{array}{ccc} \text{Hyp}(X) & \xrightarrow{q_X} & \mathbf{D}(X) \\ \text{Hyp}(f) \downarrow & & \downarrow f_* \\ \text{Hyp}(Y) & \xrightarrow{q_Y} & \mathbf{D}(Y) \end{array} \quad \text{i.e.} \quad q_Y \circ \text{Hyp}(f) = f_* \circ q_X.$$

Definition 3.2 (Structured hypotheses). Fix a set Θ of *structural descriptors*

$$\theta = (I_\theta, F_\theta, E_\theta, L_\theta, G_\theta, C_\theta) :$$

a sparse multi-index of directions, a factor graph, an evidence set, spline links, geometric data and conditional structure. Put

$$\text{Hyp}(X) = \Theta \times \mathbf{D}(X), \quad \text{Hyp}(f)(\theta, \rho) = (\theta, f_*\rho), \quad q_X(\theta, \rho) = \rho.$$

Lemma 3.3. *Hyp is a functor and q is natural.*

Proof. $\text{Hyp} = \Theta \times \mathbf{D}(-)$ is a product of a constant functor with $\mathbf{D}$, and q is the projection, which is natural because $q_Y(\text{Hyp}(f)(\theta, \rho)) = f_*\rho = f_*(q_X(\theta, \rho))$. $\square$

Remark 3.4 (Why descriptors are not transported). Directions v_j and spline links s_j are functions on X . Such data pull back along maps *into* X ; they cannot be pushed along an arbitrary $f: X \rightarrow Y$ (the expression $v_j \circ f$ is not defined), and conditional structure has no canonical pushforward along non-injective maps. In CausalIDView the descriptors parametrize a model on a fixed design, and the only datum that must move along outcome maps is the predictive distribution. [Definition 3.2](#) records exactly this. The *admissible* hypotheses $\text{Hyp}^{\text{adm}}(X) = \{(\theta, \rho_\theta)\}$, in which the predictive is the one produced by θ , form a subset that is generally not closed under $\text{Hyp}(f)$; all constructions below are carried out on the full functor Hyp .

Definition 3.5 (Shadow). $\text{Sh}(X) = \mathbf{D}(\text{Hyp}(X))$, the finite convex combinations $\Sigma = \sum_i \lambda_i \delta_{h_i}$ of hypotheses. Sh is a functor with $\text{Sh}(f) = \mathbf{D}(\text{Hyp}(f))$.

4 The structured posterior

Definition 4.1. $\text{Post}(X) = \mathbf{D}(X \times \text{Hyp}(X))$. An element is $P = \sum_k \mu_k \delta_{(x_k, h_k)}$. The *outcome marginal* is $\pi_{X*} = \mathbf{D}(\text{pr}_1): \text{Post}(X) \rightarrow \mathbf{D}(X)$, and the *predictive* is $p_X = \mu_X \circ \mathbf{D}(q_X \circ \text{pr}_2)$, so $p_X(P) = \sum_k \mu_k q_X(h_k)$.

For a nonzero finitely supported measure $m \geq 0$ write $\mathcal{N}(m) = m/\|m\|$ for its normalization. $\mathcal{N}$ absorbs positive scalars: $\mathcal{N}(cm) = \mathcal{N}(m)$ for $c > 0$.

4.1 Mixture

Definition 4.2. $P_1 \oplus_\lambda P_2 = \lambda P_1 + (1 - \lambda)P_2$ for $\lambda \in [0, 1]$.

Lemma 4.3 (Barycentric associativity). *For $\lambda, \mu \in [0, 1]$ with $\lambda\mu < 1$ and $\nu = \frac{(1-\lambda)\mu}{1-\lambda\mu}$,*

$$(P_1 \oplus_\lambda P_2) \oplus_\mu P_3 = P_1 \oplus_{\lambda\mu} (P_2 \oplus_\nu P_3).$$

If $\lambda = \mu = 1$ both sides equal P_1 for any choice of the inner weight.

Proof. The left side is $\lambda\mu P_1 + (1 - \lambda)\mu P_2 + (1 - \mu)P_3$. On the right, $(1 - \lambda\mu)\nu = (1 - \lambda)\mu$ and $(1 - \lambda\mu)(1 - \nu) = 1 - \lambda\mu - (1 - \lambda)\mu = 1 - \mu$. The boundary case is immediate. $\square$

4.2 Composition

Definition 4.4 (Compositional structure). A compositional structure on $\text{Hyp}(X)$ consists of a relation $\perp$ (compatibility), a map $\otimes$ defined on compatible pairs, and an element h_{id} , such that:

- (A1) (ternary admissibility and associativity) $h_1 \perp h_2$ and $(h_1 \otimes h_2) \perp h_3$ if and only if $h_2 \perp h_3$ and $h_1 \perp (h_2 \otimes h_3)$, and then $(h_1 \otimes h_2) \otimes h_3 = h_1 \otimes (h_2 \otimes h_3)$;
- (A2) (identity) $h_{\text{id}} \perp h$, $h \perp h_{\text{id}}$ and $h_{\text{id}} \otimes h = h \otimes h_{\text{id}} = h$ for all h .

Definition 4.5 (Composition of posteriors). The *unnormalized product* is

$$(P_1 \otimes P_2)(x, h) = \sum_{\substack{h_1 \perp h_2 \\ h_1 \otimes h_2 = h}} P_1(x, h_1) P_2(x, h_2),$$

a finitely supported measure of total mass $Z(P_1, P_2) \in [0, 1]$. If $Z(P_1, P_2) > 0$ put $P_1 \otimes P_2 = \mathcal{N}(P_1 \otimes P_2) \in \text{Post}(X)$; otherwise $P_1 \otimes P_2$ is undefined.

Remark 4.6. Without normalization the product is not a probability distribution: if $\perp$ is total its mass is $\sum_x m_1(x)m_2(x)$, where $m_i = \pi_{X*}(P_i)$, which is strictly less than 1 unless the marginals are equal point masses.

Lemma 4.7. $\otimes$ extends bilinearly to finitely supported measures and is associative. Consequently $(P_1 \otimes P_2) \otimes P_3 = P_1 \otimes (P_2 \otimes P_3)$, and either side is defined exactly when $P_1 \otimes P_2 \otimes P_3 \neq 0$.

Proof. Bilinearity is clear from the formula. Expanding both triple products gives sums over triples with $(h_1 \otimes h_2) \otimes h_3 = h$, respectively $h_1 \otimes (h_2 \otimes h_3) = h$, over the admissible triples; by (A1) these index sets and terms coincide. Since $\otimes$ is bilinear and $\mathcal{N}$ absorbs positive scalars, $\mathcal{N}(\mathcal{N}(m) \otimes m') = \mathcal{N}(m \otimes m')$, so both normalized products equal $\mathcal{N}(P_1 \otimes P_2 \otimes P_3)$. If $P_1 \otimes P_2 = 0$ the triple product is 0, which gives the definedness statement. $\square$

Proposition 4.8 (Distributivity). Let $Z_{i3} = Z(P_i, P_3)$ and suppose $Z_{13}, Z_{23} > 0$. Then

$$(P_1 \oplus_\lambda P_2) \otimes P_3 = (P_1 \otimes P_3) \oplus_{\lambda''} (P_2 \otimes P_3), \quad \lambda'' = \frac{\lambda Z_{13}}{\lambda Z_{13} + (1 - \lambda) Z_{23}}.$$

Proof. By bilinearity $(\lambda P_1 + (1 - \lambda) P_2) \otimes P_3 = \lambda Z_{13} (P_1 \otimes P_3) + (1 - \lambda) Z_{23} (P_2 \otimes P_3)$; normalize. $\square$

The weight is $\lambda'' = \lambda$ only when $Z_{13} = Z_{23}$ (or $\lambda \in \{0, 1\}$).

4.3 Conditioning

Definition 4.9. For evidence e with likelihood $L(e \mid \cdot) : \text{Hyp}(X) \rightarrow [0, \infty)$, put $Z_e(P) = \sum_{(x, h)} L(e \mid h) P(x, h)$ and, when $Z_e(P) > 0$, $\text{Cond}_e(P) = \mathcal{N}(L(e \mid \cdot) P)$.

Proposition 4.10 (Chain rule). Suppose $L(e_1 \wedge e_2 \mid h) = L(e_1 \mid h) L(e_2 \mid h, e_1)$, and write $\text{Cond}_{e_2|e_1}$ for conditioning with likelihood $L(e_2 \mid \cdot, e_1)$. Then $\text{Cond}_{e_2|e_1}(\text{Cond}_{e_1} P) = \text{Cond}_{e_1 \wedge e_2}(P)$ whenever the right side is defined. In particular, if e_1, e_2 are conditionally independent given h , then $\text{Cond}_{e_2} \circ \text{Cond}_{e_1} = \text{Cond}_{e_1 \wedge e_2}$.

Proof. $\mathcal{N}(L_2 \cdot \mathcal{N}(L_1 P)) = \mathcal{N}(L_2 L_1 P)$. $\square$

Proposition 4.11 (Conditioning and mixture). If $Z_1 = Z_e(P_1) > 0$ and $Z_2 = Z_e(P_2) > 0$, then $\text{Cond}_e(P_1 \oplus_\lambda P_2) = \text{Cond}_e(P_1) \oplus_{\lambda'} \text{Cond}_e(P_2)$ with $\lambda' = \lambda Z_1 / (\lambda Z_1 + (1 - \lambda) Z_2)$.

Proof. $L \cdot (\lambda P_1 + (1 - \lambda)P_2) = \lambda Z_1 \text{Cond}_e(P_1) + (1 - \lambda)Z_2 \text{Cond}_e(P_2)$; normalize. $\square$

Proposition 4.12 (Conditioning and composition). *If $L(e \mid h_1 \otimes h_2) = L(e \mid h_1)L(e \mid h_2)$ for all compatible pairs, then $\text{Cond}_e(P_1 \otimes P_2) = \text{Cond}_e(P_1) \otimes \text{Cond}_e(P_2)$, and the left side is defined exactly when the right side is.*

Proof. Pointwise, $L(e \mid h_1 \otimes h_2)P_1(x, h_1)P_2(x, h_2) = (LP_1)(x, h_1)(LP_2)(x, h_2)$, so $L \cdot (P_1 \otimes P_2) = (LP_1) \otimes (LP_2)$. Using that $\mathcal{N}$ absorbs positive scalars and $\otimes$ is bilinear, $\text{Cond}_e(P_1 \otimes P_2) = \mathcal{N}(L \cdot (P_1 \otimes P_2)) = \mathcal{N}((LP_1) \otimes (LP_2)) = \mathcal{N}(\text{Cond}_e P_1 \otimes \text{Cond}_e P_2)$. Both sides are defined iff $(LP_1) \otimes (LP_2) \neq 0$. $\square$

Remark 4.13 (Partial algebra). $\text{Post}(X)$ is a convex set with a partial, normalized, associative composition and a partial conditioning action. It is not a semiring: $\oplus$ distributes over $\otimes$ only with the reweighting of [proposition 4.8](#).

5 Observational equivalence

Definition 5.1. $P_1 \equiv P_2$ iff $\pi_{X*}(P_1) = \pi_{X*}(P_2)$.

Theorem 5.2 (Affine first isomorphism theorem). *π_{X*} is affine. The relation $\equiv$ is a congruence for convex combinations, so $\text{Post}(X)/\equiv$ carries a unique convex structure making the quotient map affine, and π_{X*} induces an affine injection $\text{Post}(X)/\equiv \hookrightarrow \text{D}(X)$. If $\text{Hyp}(X) \neq \emptyset$ it is a bijection, so $\text{Post}(X)/\equiv \cong \text{D}(X)$ as convex sets.*

Proof. $\pi_{X*} = \text{D}(\text{pr}_1)$ is affine. If $P_i \equiv P'_i$ then $\pi_{X*}(\sum a_i P_i) = \sum a_i \pi_{X*}(P_i) = \pi_{X*}(\sum a_i P'_i)$, so $\equiv$ is a congruence and the induced map is well defined, affine and injective. For surjectivity pick $h_0 \in \text{Hyp}(X)$; then $\pi_{X*}(\sum_x \rho(x) \delta_{(x, h_0)}) = \rho$. $\square$

Proposition 5.3 (Congruence for $\oplus$). *If $P_1 \equiv P'_1$ and $P_2 \equiv P'_2$ then $P_1 \oplus_\lambda P_2 \equiv P'_1 \oplus_\lambda P'_2$.*

Proposition 5.4 (Congruence for $\otimes$). *If $\perp$ is total, then $\pi_{X*}(P_1 \otimes P_2) = \mathcal{N}(m_1 m_2)$ with $m_i = \pi_{X*}(P_i)$, so $\equiv$ is a congruence for $\otimes$ (no condition on q is needed). If $\perp$ is not total this can fail.*

Proof. With $\perp$ total, $\sum_h (P_1 \otimes P_2)(x, h) = \sum_{h_1, h_2} P_1(x, h_1)P_2(x, h_2) = m_1(x)m_2(x)$. For the failure, take $X = \{x, y\}$, hypotheses h_1, h_2, h_3 with $h_1 \perp h_3$ but not $h_2 \perp h_3$, and $P_1 = \frac{1}{2}\delta_{(x, h_1)} + \frac{1}{2}\delta_{(y, h_2)}$, $P'_1 = \frac{1}{2}\delta_{(x, h_2)} + \frac{1}{2}\delta_{(y, h_1)}$, $P_2 = \frac{1}{2}\delta_{(x, h_3)} + \frac{1}{2}\delta_{(y, h_3)}$. Then $P_1 \equiv P'_1$ but the marginals of $P_1 \otimes P_2$ and $P'_1 \otimes P_2$ are δ_x and δ_y . $\square$

Proposition 5.5 (Congruence for Cond). *Let $|X| \geq 2$. Then $\equiv$ is a congruence for Cond_e (on all P where both sides are defined) if and only if $L(e \mid \cdot)$ is constant on $\text{Hyp}(X)$.*

Proof. If L is constant and positive, $\text{Cond}_e = \text{id}$. Conversely, suppose $L(e \mid h_1) = \ell_1 \neq \ell_2 = L(e \mid h_2)$, so $\ell_1 + \ell_2 > 0$. With distinct $x_1, x_2 \in X$ let $P = \frac{1}{2}\delta_{(x_1, h_1)} + \frac{1}{2}\delta_{(x_2, h_2)}$ and $P' = \frac{1}{2}\delta_{(x_1, h_2)} + \frac{1}{2}\delta_{(x_2, h_1)}$. Then $P \equiv P'$, but the marginals of $\text{Cond}_e P$ and $\text{Cond}_e P'$ are $(\ell_1, \ell_2)/(\ell_1 + \ell_2)$ and $(\ell_2, \ell_1)/(\ell_1 + \ell_2)$, which differ. $\square$

Remark 5.6. In particular, requiring $L(e \mid h) = \ell(e \mid q_X(h))$ does *not* make $\equiv$ a congruence for conditioning: in the counterexample $q_X(h_1) \neq q_X(h_2)$ is allowed. With $\ell_1 = 2, \ell_2 = 1$ the conditioned marginals are $(\frac{2}{3}, \frac{1}{3})$ and $(\frac{1}{3}, \frac{2}{3})$.

6 The three-level tower and the context comonad

6.1 The three posterior levels

For a set X let $H_3(X) = D(X) \times D^2(X) \times D^3(X)$, with elements (ρ, Π, Ω) . The barycentric projections $b_X = \mu_X: D^2X \rightarrow DX$ and $b_X^{(2)} = \mu_{DX}: D^3X \rightarrow D^2X$ form the tower $D^3X \rightarrow D^2X \rightarrow DX$. By monad associativity $\mu_X \circ \mu_{DX} = \mu_X \circ D(\mu_X)$, so the full collapse $D^3X \rightarrow DX$ does not depend on which projection is used at the top. The three levels are distinct state; the projections are observation maps.

6.2 A general context comonad

For a set X , an element c of a set A , and $y \in X$, write $j_{X,c}(y) = (y, c)$.

Theorem 6.1 (Context comonad). *Let $E: \mathbf{Set} \rightarrow \mathbf{Set}$ be any functor. Put $C_E(X) = X \times E(X)$ and $C_E(f) = f \times E(f)$, and define*

$$\varepsilon_X(x, c) = x, \quad \delta_X(x, c) = ((x, c), \kappa_X(c)), \quad \kappa_X(c) = E(j_{X,c})(c) \in E(C_EX),$$

where $j_{X,c}: X \rightarrow C_EX$. Then $(C_E, \varepsilon, \delta)$ is a comonad on $\mathbf{Set}$.

Proof. Write $C = C_E$. Functoriality is that of $\text{id} \times E$, and ε is natural because it is a product projection.

Naturality of δ . For $f: X \rightarrow Y$ we need $\delta_Y(fx, Ef c) = CCf(\delta_X(x, c))$. The second component of the right side is $E(Cf)(E(j_{X,c})(c)) = E(Cf \circ j_{X,c})(c)$. Now $(Cf \circ j_{X,c})(y) = (fy, Ef c) = (j_{Y, Ef c} \circ f)(y)$, so this equals $E(j_{Y, Ef c})(Ef c) = \kappa_Y(Ef c)$. First components agree trivially.

Counit laws. $\varepsilon_{CX} \delta_X(x, c) = (x, c)$. Also $C(\varepsilon_X) \delta_X(x, c) = (\varepsilon_X(x, c), E(\varepsilon_X)E(j_{X,c})(c)) = (x, c)$ since $\varepsilon_X \circ j_{X,c} = \text{id}_X$.

Coassociativity. Let $w = (x, c)$ and $c' = \kappa_X(c)$. We first note

$$\delta_X \circ j_{X,c} = j_{CX,c'} \circ j_{X,c}, \tag{1}$$

since both send y to $((y, c), c')$; this uses that $\kappa_X(c)$ depends only on c . Now $\delta_{CX} \delta_X(x, c) = ((w, c'), E(j_{CX,c'})(c')) = ((w, c'), E(j_{CX,c'} \circ j_{X,c})(c))$, while $C(\delta_X) \delta_X(x, c) = (\delta_X(w), E(\delta_X)(c')) = ((w, c'), E(\delta_X \circ j_{X,c})(c))$. These agree by (1). $\square$

Remark 6.2. If E is constant at a set A , then C_E is the reader (product) comonad $X \times A$. For general E the context is transported into C_EX by κ_X rather than copied.

6.3 The three-level context

Definition 6.3. $E_3(X) = D(X) \times D^2(X) \times D^3(X) \times \text{Sh}(X)$ and $C_3 = C_{E_3}$, so $C_3(X) = X \times DX \times D^2X \times D^3X \times \text{Sh}X$ with points $(x, \rho, \Pi, \Omega, \Sigma)$.

Corollary 6.4. C_3 is a comonad. Explicitly, for $c = (\rho, \Pi, \Omega, \Sigma)$,

$$\kappa_X(c) = ((j_{X,c})_* \rho, D^2(j_{X,c})(\Pi), D^3(j_{X,c})(\Omega), D(\text{Hyp}(j_{X,c}))(\Sigma)).$$

Each component of $E_3(Y)$ is of the form $D(Z)$, so $E_3(Y)$ is a convex set, and for every f the map $E_3(f)$ is affine.

6.4 The context lift

Definition 6.5. $J_{X,c} = \text{Hyp}(j_{X,c}) : \text{Hyp}(X) \rightarrow \text{Hyp}(\mathbf{C}_3 X)$; explicitly $J_{X,c}(\theta, \rho) = (\theta, (j_{X,c})_* \rho)$.

Proposition 6.6 (Coherence of J). (CL1) $q_{\mathbf{C}_3 X} \circ J_{X,c} = (j_{X,c})_* \circ q_X$;

(CL2) $\text{Hyp}(\varepsilon_X) \circ J_{X,c} = \text{id}_{\text{Hyp}(X)}$;

(CL3) $\text{Hyp}(\delta_X) \circ J_{X,c} = J_{\mathbf{C}_3 X, \kappa_X(c)} \circ J_{X,c}$.
The shadow component of $\kappa_X(c)$ is $(J_{X,c})_* \Sigma$.

Proof. (CL1) is naturality of q . (CL2) is functoriality of Hyp and $\varepsilon_X \circ j_{X,c} = \text{id}$. (CL3) is functoriality of Hyp applied to (1). $\square$

7 The distribution-level lift

For $k \geq 1$ let $\eta_X^{(k)} = \eta_{\mathbf{D}^k X} \circ \cdots \circ \eta_{\mathbf{D} X} : \mathbf{D} X \rightarrow \mathbf{D}^{k+1} X$, so $\eta_X^{(3)}(\rho) = \delta_{\delta_{\delta_\rho}}$.

Definition 7.1. $K_X^{[k]} : \text{Hyp}(X) \rightarrow \text{Hyp}(\mathbf{D}^k X)$, $K_X^{[k]}(\theta, \rho) = (\theta, \eta_X^{(k)}(\rho))$. Then $q_{\mathbf{D}^k X}(K_X^{[k]} h) = \eta_X^{(k)}(q_X h)$; for $k = 3$ this is $\delta_{\delta_{\delta_{\rho_h}}}$.

Proposition 7.2 (Naturality and K/J coherence). $K^{[k]} : \text{Hyp} \Rightarrow \text{Hyp} \circ \mathbf{D}^k$ is natural. In particular

$$\text{Hyp}(\mathbf{D}^3 j_{X,c}) \circ K_X^{[3]} = K_{\mathbf{C}_3 X}^{[3]} \circ J_{X,c}.$$

Proof. $\eta^{(k)}$ is a composite of whiskered units, hence natural: $\mathbf{D}^{k+1}(f) \circ \eta_X^{(k)} = \eta_Y^{(k)} \circ \mathbf{D}(f)$. Apply with $f = j_{X,c}$. $\square$

Remark 7.3. The coherence must use $\mathbf{D}^3 j_{X,c}$: $K_X^{[3]}$ lands in $\text{Hyp}(\mathbf{D}^3 X)$, on which $\text{Hyp}(\mathbf{D}^2 j)$ does not act. Note also that $K_X^{[3]}(h)$ is a point mass at the single point $\delta_{\delta_{\rho_h}}$ of $\mathbf{D}^3 X$; it is not the pushforward of ρ_h along $x \mapsto \delta_{\delta_x}$, which would be $\text{Hyp}(\eta\eta\eta)$.

8 Mixed distributive laws for the tower

Throughout, $M = \sum_i a_i \delta_{(x_i, c_i)} \in \mathbf{DC}_3 X$ with $c_i = (\rho_i, \Pi_i, \Omega_i, \Sigma_i)$. The target is $\mathbf{C}_3 \mathbf{D} X = \mathbf{D} X \times \mathbf{D}(\mathbf{D} X) \times \mathbf{D}^2(\mathbf{D} X) \times \mathbf{D}^3(\mathbf{D} X) \times \text{Sh}(\mathbf{D} X)$.

8.1 What the unit axiom forces

Lemma 8.1. If $\lambda : \mathbf{DC}_3 \Rightarrow \mathbf{C}_3 \mathbf{D}$ satisfies (U), then $\lambda(\delta_{(x,c)}) = (\delta_x, \mathbf{E}_3(\eta_X)(c))$. In particular its level-one component is $\mathbf{D}(\eta_X)(\rho) = \sum_y \rho(y) \delta_{\delta_y}$, which differs from δ_ρ whenever ρ is not a point mass. The two agree after barycentric collapse: $\mu_X(\mathbf{D}(\eta_X)\rho) = \rho = \mu_X(\delta_\rho)$.

Proof. (U) states $\lambda(\eta_{\mathbf{C}_3 X}(x, c)) = \mathbf{C}_3(\eta_X)(x, c) = (\eta_X x, \mathbf{E}_3(\eta_X)c)$. $\square$

So a law that keeps a non-degenerate posterior ρ as a single hypothesis δ_ρ at level one cannot satisfy (U); the unit axiom requires level-one information to be spread over point hypotheses. This is the precise form of the tension between hierarchy and barycentric collapse.

8.2 The shift law

Definition 8.2 (Shift law). With $K_X^{[1]}(\theta, \rho) = (\theta, \delta_\rho)$,

$$\lambda_X^\sigma(M) = \left(\sum_i a_i \delta_{x_i}, \sum_i a_i \delta_{\rho_i}, \sum_i a_i \delta_{\Pi_i}, \sum_i a_i \delta_{\Omega_i}, \sum_i a_i (K_X^{[1]})_* \Sigma_i \right).$$

The components lie in $\mathbf{D}X$, $\mathbf{D}(\mathbf{D}X)$, $\mathbf{D}(\mathbf{D}^2 X) = \mathbf{D}^2(\mathbf{D}X)$, $\mathbf{D}(\mathbf{D}^3 X) = \mathbf{D}^3(\mathbf{D}X)$ and $\mathbf{D}(\mathbf{HypD}X) = \mathbf{Sh}(\mathbf{D}X)$ respectively, so $\lambda_X^\sigma: \mathbf{DC}_3 X \rightarrow \mathbf{C}_3 \mathbf{D}X$.

Theorem 8.3. λ^σ is natural and satisfies (Co). It violates (U), (M) and (Cm). The defects in (U) and (M) at level one vanish after applying μ_X .

Proof. Naturality. Each component of λ^σ is the pushforward of M along a natural projection (followed, for the shadow, by the natural map $K^{[1]}$ and barycenter, both natural).

(Co). $\varepsilon_{\mathbf{D}X}(\lambda^\sigma M) = \sum_i a_i \delta_{x_i} = \mathbf{D}(\varepsilon_X)(M)$.

(U) fails. By lemma 8.1: $\lambda^\sigma(\delta_{(x,c)})$ has level one δ_ρ , while $\mathbf{C}_3(\eta_X)(x, c)$ has $\mathbf{D}(\eta_X)\rho$. For $X = \{0, 1\}$ and $\rho = \frac{1}{2}\delta_0 + \frac{1}{2}\delta_1$ these are δ_ρ and $\frac{1}{2}\delta_{\delta_0} + \frac{1}{2}\delta_{\delta_1}$.

(M) fails. Let $M' = \sum_\ell \nu_\ell \delta_{M_\ell}$ with $M_\ell = \sum_k a_{\ell k} \delta_{(x_{\ell k}, c_{\ell k})}$. The level-one component of $\lambda^\sigma \mu(M')$ is $\sum_{\ell, k} \nu_\ell a_{\ell k} \delta_{\rho_{\ell k}}$. On the other side, $\lambda_{\mathbf{D}X}^\sigma$ reads the level-one components $\sum_k a_{\ell k} \delta_{\rho_{\ell k}}$ of $\lambda^\sigma(M_\ell)$ and $\mathbf{C}_3(\mu_X)$ applies $\mathbf{D}(\mu_X)$, giving $\sum_\ell \nu_\ell \delta_{\sum_k a_{\ell k} \rho_{\ell k}}$. These differ whenever the $\rho_{\ell k}$ for fixed ℓ are not all equal, and both have barycenter $\sum_{\ell, k} \nu_\ell a_{\ell k} \rho_{\ell k}$.

(Cm) fails. Take $M = \frac{1}{2}\delta_{(x, c_1)} + \frac{1}{2}\delta_{(x, c_2)}$ with $\rho_1 \neq \rho_2$. The level-one component of $\delta_{\mathbf{D}X}(\lambda^\sigma M)$ is $\mathbf{D}(j_{\mathbf{D}X, e})(\sum_i \frac{1}{2}\delta_{\rho_i}) = \sum_i \frac{1}{2}\delta_{(\rho_i, e)}$, where e is the pooled context of $\lambda^\sigma M$. On the other side, $\mathbf{C}_3(\lambda^\sigma) \circ \lambda_{\mathbf{C}_3 X}^\sigma \circ \mathbf{D}(\delta_X)$ gives $\sum_i \frac{1}{2}\delta_{\lambda^\sigma((j_{X, c_i})_* \rho_i)} = \sum_i \frac{1}{2}\delta_{(\rho_i, e_i)}$ with $e_i = (\delta_{\rho_i}, \delta_{\Pi_i}, \delta_{\Omega_i}, (K^{[1]})_* \Sigma_i)$, and $e_i \neq e$. $\square$

Remark 8.4. The transformation $\lambda^{[3]}$ of the earlier version is λ^σ with the level-three component omitted and a shadow component of the wrong type. Its claimed unit and comultiplication theorems fail by theorem 8.3.

8.3 The averaging law

Let $\beta: \mathbf{D}(\mathbf{E}_3 Y) \rightarrow \mathbf{E}_3 Y$ be the barycenter, computed componentwise by μ on each factor $\mathbf{D}(Z)$; every $\mathbf{E}_3(f)$ commutes with β because it is affine.

Definition 8.5 (Averaging law).

$$\bar{\lambda}_X(M) = \left(\sum_i a_i \delta_{x_i}, \sum_i a_i \mathbf{E}_3(\eta_X)(c_i) \right),$$

i.e. $\bar{\lambda}_X = \langle \mathbf{D}(\varepsilon_X), \beta \circ \mathbf{D}(\mathbf{E}_3(\eta_X) \circ \text{pr}_2) \rangle$.

Theorem 8.6. $\bar{\lambda}$ is natural and satisfies (U), (Co) and (M) exactly.

Proof. Naturality. For $f: X \rightarrow Y$, using that $\mathbf{E}_3(\mathbf{D}f)$ is affine and $\mathbf{D}f \circ \eta_X = \eta_Y \circ f$,

$$\mathbf{E}_3(\mathbf{D}f) \left(\sum_i a_i \mathbf{E}_3(\eta_X) c_i \right) = \sum_i a_i \mathbf{E}_3(\eta_Y)(\mathbf{E}_3(f) c_i),$$

which is the context component of $\bar{\lambda}_Y(\mathbf{D}(\mathbf{C}_3 f)M)$; first components agree by naturality of $\mathbf{D}(\varepsilon)$.

(U). $\bar{\lambda}(\delta_{(x,c)}) = (\delta_x, \mathbf{E}_3(\eta_X)c) = \mathbf{C}_3(\eta_X)(x, c)$.

	natural	(U)	(Co)	(M)	(Cm)	keeps δ_ρ at level one
shift law λ^σ	yes	no	yes	no	no	yes
averaging law $\bar{\lambda}$	yes	yes	yes	yes	no	no

Table 1: Status of the two mixed distributive laws $\text{DC}_3 \Rightarrow \text{C}_3\text{D}$ (theorems 8.3, 8.6 and 8.7). By lemma 8.1, no law satisfying (U) keeps δ_ρ at level one.

(Co). $\varepsilon_{\text{DX}} \circ \bar{\lambda}_X = \text{D}(\varepsilon_X)$ by definition.

(M). Let $M' = \sum_\ell \nu_\ell \delta_{M_\ell}$ with $M_\ell = \sum_k a_{\ell k} \delta_{(x_{\ell k}, c_{\ell k})}$, and put

$$P_\ell = \sum_k a_{\ell k} \delta_{x_{\ell k}}, \quad e_\ell = \sum_k a_{\ell k} \text{E}_3(\eta_X) c_{\ell k}.$$

The left side is $(\sum_\ell \nu_\ell P_\ell, \sum_\ell \nu_\ell e_\ell)$. On the right, $\text{D}(\bar{\lambda})M' = \sum_\ell \nu_\ell \delta_{(P_\ell, e_\ell)}$; then $\bar{\lambda}_{\text{DX}}$ gives

$$\left(\sum_\ell \nu_\ell \delta_{P_\ell}, \sum_\ell \nu_\ell \text{E}_3(\eta_{\text{DX}}) e_\ell \right),$$

and $\text{C}_3(\mu_X)$ turns this into

$$\left(\sum_\ell \nu_\ell P_\ell, \sum_\ell \nu_\ell \text{E}_3(\mu_X \circ \eta_{\text{DX}}) e_\ell \right) = \left(\sum_\ell \nu_\ell P_\ell, \sum_\ell \nu_\ell e_\ell \right),$$

using that $\text{E}_3(\mu_X)$ is affine and $\mu_X \circ \eta_{\text{DX}} = \text{id}$. $\square$

Theorem 8.7. $\bar{\lambda}$ violates (Cm).

Proof. Let $X = \{0, 1\}$, fix Π, Ω, Σ , and let $c_i = (\delta_i, \Pi, \Omega, \Sigma)$ for $i = 0, 1$, $M = \frac{1}{2}\delta_{(0, c_0)} + \frac{1}{2}\delta_{(0, c_1)}$. Write $e_i = \text{E}_3(\eta_X) c_i$ and $e = \frac{1}{2}e_0 + \frac{1}{2}e_1$, so $\bar{\lambda}M = (\delta_0, e)$ and the level-one component of e is $\frac{1}{2}\delta_{\delta_0} + \frac{1}{2}\delta_{\delta_1}$. The level-one component of $\delta_{\text{DX}}(\bar{\lambda}M)$ is $\text{D}(j_{\text{DX}, e})$ of that, namely $\frac{1}{2}\delta_{(\delta_0, e)} + \frac{1}{2}\delta_{(\delta_1, e)}$.

On the other side, $\text{D}(\delta_X)M = \sum_i \frac{1}{2}\delta_{((0, c_i), \kappa_X(c_i))}$. Since $\text{E}_3(\bar{\lambda}_X)$ is affine and $\bar{\lambda}_X \circ \eta_{\text{C}_3 X} \circ j_{X, c_i} = j_{\text{DX}, e_i} \circ \eta_X$ (by (U)), the context component of $\text{C}_3(\bar{\lambda})\bar{\lambda}_{\text{C}_3 X} \text{D}(\delta_X)(M)$ is $\sum_i \frac{1}{2}\text{E}_3(j_{\text{DX}, e_i})(e_i)$, whose level-one component is $\frac{1}{2}\delta_{(\delta_0, e_0)} + \frac{1}{2}\delta_{(\delta_1, e_1)}$. Since $e_0 \neq e \neq e_1$, the two sides differ. $\square$

Remark 8.8. In general the context component of the right side of (Cm) is $\sum_i a_i \text{E}_3(j_{\text{DX}, e_i})(e_i)$ and that of the left side is $\text{E}_3(j_{\text{DX}, e})(e)$ with $e = \sum_i a_i e_i$: comultiplication copies a *single* pooled context into C_3DX , whereas the right side keeps one context per sample. Both sides do agree after $\text{C}_3(\varepsilon_{\text{DX}})$ and after $\varepsilon_{\text{C}_3\text{DX}}$, but this follows from (Co) and naturality alone and carries no extra information.

Remark 8.9 (Relation to weak distributive laws). The weak distributive laws of Street and Böhm [21, 10, 11], used by Garner [15] and Goy–Petrişan [16], are laws between two monads that keep the multiplication axioms and weaken the unit axioms; the composite is obtained by splitting an idempotent. Neither λ^σ nor $\bar{\lambda}$ falls under that notion as stated: λ^σ fails a multiplication-type axiom (M), and $\bar{\lambda}$ fails the comonad analogue (Cm). A formulation in which a failing axiom is witnessed by a 2-cell would need a 2-categorical structure on the base (for instance an order or convex enrichment of the relevant hom-sets); on **Set** itself the only 2-cells between natural transformations are identities, so no non-identity 2-cell α is available. The earlier version’s claim of a weak distributive law with a barycentric 2-cell is therefore withdrawn.

9 Nondegeneracy and the likelihood-factorization criterion

Axiom 9.1 (Nondegeneracy). (a) $L(e \mid h_{\text{id}}) = 1$ for all e ; (b) $L(e \mid h) > 0$ for all h, e .

Remark 9.2 (Composite identifiability is too strong). The earlier version assumed that $\otimes$ is injective on pairs: $h_1 \otimes h_2 = h'_1 \otimes h'_2 \Rightarrow (h_1, h_2) = (h'_1, h'_2)$. Together with (A2) this forces $\text{Hyp}(X) = \{h_{\text{id}}\}$: for any h , $h \otimes h_{\text{id}} = h = h_{\text{id}} \otimes h$, so $(h, h_{\text{id}}) = (h_{\text{id}}, h)$ and $h = h_{\text{id}}$. Restricting to supports does not help when the conclusion is quantified over all posteriors.

Theorem 9.3 (The “iff”). Assume [axiom 9.1](#). Then $\text{Cond}_e(P_1 \otimes P_2) = \text{Cond}_e(P_1) \otimes \text{Cond}_e(P_2)$ for all P_1, P_2 (whenever either side is defined) if and only if $L(e \mid h_1 \otimes h_2) = L(e \mid h_1)L(e \mid h_2)$ for all compatible pairs $h_1 \perp h_2$.

Proof. ($\Leftarrow$) is [proposition 4.12](#). ($\Rightarrow$) Let $h_1 \perp h_2$. If $h_1 = h_2 = h_{\text{id}}$ there is nothing to prove, since $L(e \mid h_{\text{id}}) = 1$. Otherwise fix $x \in X$ and $a, b \in (0, 1)$ and set $P_1 = a\delta_{(x, h_1)} + (1 - a)\delta_{(x, h_{\text{id}})}$, $P_2 = b\delta_{(x, h_2)} + (1 - b)\delta_{(x, h_{\text{id}})}$. Write $\alpha(h_1) = a$, $\alpha(h_{\text{id}}) = 1 - a$, $\beta(h_2) = b$, $\beta(h_{\text{id}}) = 1 - b$. By (A2) every pair from $\{h_1, h_{\text{id}}\} \times \{h_2, h_{\text{id}}\}$ is compatible, and since all mass sits on one outcome, $P_1 \otimes P_2$ has mass 1. For each $k \in \text{Hyp}(X)$ put

$$w(k) = \sum_{u \otimes v = k} \alpha(u)\beta(v) L(e \mid k), \quad w'(k) = \sum_{u \otimes v = k} \alpha(u)\beta(v) L(e \mid u)L(e \mid v),$$

summing over $(u, v) \in \{h_1, h_{\text{id}}\} \times \{h_2, h_{\text{id}}\}$. Then $\text{Cond}_e(P_1 \otimes P_2) = \mathcal{N}(w)$ and, using that all mass is on one outcome, $\text{Cond}_e P_1 \otimes \text{Cond}_e P_2 = \mathcal{N}(w')$. For pairs involving h_{id} , $L(e \mid u \otimes h_{\text{id}}) = L(e \mid u) = L(e \mid u)L(e \mid h_{\text{id}})$ by (A2) and [axiom 9.1\(a\)](#), so

$$w - w' = ab (L(e \mid h_1 \otimes h_2) - L(e \mid h_1)L(e \mid h_2)) \delta_{h_1 \otimes h_2}.$$

The composites $h_1 \otimes h_2$, h_1 , h_2 , h_{id} are not all equal (else $h_1 = h_2 = h_{\text{id}}$), so there is $k_0 \neq h_1 \otimes h_2$ among them, and $w(k_0) = w'(k_0) > 0$ by [axiom 9.1\(b\)](#). Equality $\mathcal{N}(w) = \mathcal{N}(w')$ means $w = cw'$ for some $c > 0$; evaluating at k_0 gives $c = 1$, hence $w = w'$ and $L(e \mid h_1 \otimes h_2) = L(e \mid h_1)L(e \mid h_2)$. $\square$

Remark 9.4. No identifiability or separation assumption is needed: the hypothesis h_{id} already supplies a reference point that fixes the normalizing constant. The earlier proof’s constant $c = Z/(Z_1 Z_2)$ depended on the chosen posteriors and could not be set to 1 by a separate choice.

10 Empirical setup

The empirical setting is the CausalIDView framework for gradient-free causal identification with structured shadows [\[3\]](#). The outcome space X consists of binary causal identification outcomes; descriptors θ are sparse multi-index hypotheses with spline links, nominally $r = 3$ directions; the posterior is sampled by gradient-free Metropolis. Targets are Manski-type partial-identification bounds [\[18\]](#) and instrumental-variable bounds.

On the 40-world benchmark ([appendix A](#)), the posterior under a generic prior with 16 pooled chains attains Manski RMSE 0.0934 and IV RMSE 0.1226, against 0.105 and 0.138 for the previous structured backbone. These numbers are close to the approximate TabPFN reference values [\[17\]](#); since TabPFN was not rerun under the same protocol and no uncertainty intervals are reported, we do not claim parity. The flattening of the chain-pooling curve is consistent with, but does not by itself test, saturation of the level-two hierarchy. The 90% bootstrap intervals cover the true bounds only 70–71% of the time, which indicates substantial miscalibration; bias-aware intervals remain open.

The categorical results above do not depend on these experiments, and the experiments do not test the categorical results.

11 Related work

Three-layer realisation, hierarchy and holonomy. [6, 4, 5].

Categorical probability. The finite distribution monad and Markov categories [12, 14]; a synthetic de Finetti theorem [13].

Distributive laws. Beck’s distributive laws [9]; the formal theory of monads [20]; mixed laws between a monad and a comonad [19]; weak distributive laws [21, 10, 11], with applications to the Vietoris monad [15] and to combining probabilistic and nondeterministic choice [16].

Information cohomology and contextuality. Baudot–Bennequin [8], Vigneaux [22], Abramsky–Brandenburger [1] and Abramsky–Mansfield–Barbosa [2].

Empirical. CausalIDView [3]; TabPFN [17]; partial identification [18]; spiked random-matrix thresholds [7].

12 Summary of results

1. $C_E(X) = X \times E(X)$ is a comonad for every E ; C_3 is an instance (proved).
2. (Hyp, q) with q natural; $J = \text{Hyp}(j)$ satisfies CL1–CL3; $K^{[k]}$ is natural and coheres with J via D^3j (proved).
3. Normalized partial algebra on $\text{Post}(X)$ with reweighted distributivity and conditioning laws (proved).
4. Affine first isomorphism theorem; congruence for $\otimes$ when $\perp$ is total; congruence for Conde iff $L(e \mid \cdot)$ is constant (proved, with counterexamples).
5. Shift law: natural and (Co); fails (U), (M), (Cm). Averaging law: natural, (U), (Co), (M); fails (Cm). The unit axiom excludes point-mass hierarchy at level one (proved).
6. Composite identifiability trivializes Hyp; the “iff” holds under nondegeneracy alone (proved).

13 Open problems

1. Does any mixed distributive law $DC_3 \Rightarrow C_3D$ satisfy all four axioms of [definition 2.1](#)? Equivalently, does D lift to the Eilenberg–Moore category of C_3 ?
2. Is there an enrichment of the base (order, convex or metric) in which the defects of λ^σ or $\bar{\lambda}$ are witnessed by coherent 2-cells, giving a genuine weak or lax law?
3. Weaken [axiom 9.1](#) in [theorem 9.3](#), e.g. allow zero likelihoods.
4. Extension to non-discrete outcome spaces (Giry monad).
5. Calibrated uncertainty and a test that could falsify the level-two saturation interpretation of the chain-pooling curve.

14 Conclusion

The three-level tower carries a genuine comonad, and its hypothesis lifts are coherent by functoriality. The interaction with the probability monad is subtler than previously claimed: the hierarchy-preserving shift law fails three Beck axioms, while the averaging law satisfies all but comultiplication at the cost of expressing hierarchy through $D(\eta)$. The unit axiom alone rules out point-mass hierarchy at level one. Whether a full mixed distributive law exists for C_3 is open.

A Empirical tables

A.1 The 40-world benchmark

All numbers are on the 40-world set with 30 bootstrap refits per world.

Method	Manski RMSE	IV RMSE
Logistic regression (rebuild)	0.132	0.169
Structured backbone (previous best)	0.105	0.138
Posterior under generic prior, 8 chains	0.0937	0.1230
Posterior under generic prior, 16 chains	0.0934	0.1226
TabPFN, approximate reference [†]	≈ 0.09	≈ 0.12
Constant predictor	0.131	—

[†]Approximate values, not produced under this benchmark’s protocol; see [appendix B](#), item 13.

A.2 Chain-pooling curve

Chains	1	2	4	8	16
Manski	0.1005	0.0973	0.0953	0.0937	0.0934
IV	0.1319	0.1277	0.1250	0.1230	0.1226

Differences between 8 and 16 chains (about 3×10^{-4}) are reported without standard errors and should not be over-interpreted.

A.3 Coverage

The 90% bootstrap intervals cover the true bounds 70–71% of the time (82% before the more regularized backbone).

A.4 Shallow-MLP worlds

On shallow-MLP worlds the structured backbone scores 0.104 against 0.103 for the previous best, i.e. no improvement. A misspecified level-two shadow is one possible explanation; it has not been tested.

A.5 Spectral pre-pass

The pre-pass recovers on average 2.5 of 6.6 planted directions and 4.5 of 12 active coordinates per direction. In spiked random-matrix models, the Baik–Ben Arous–Péché transition [7] gives a signal-strength threshold below which the top sample eigenvector carries no information about a planted direction. Whether the CausalIDView pre-pass operates in such a regime has not been established, so we state the low recovery rate without attributing it to that threshold.

B Changes from the previous version

1. Base category changed from finite sets to **Set**, since $D(X)$ is infinite.
2. Hyp redefined as $\Theta \times D(-)$; transport of directions and splines by $v_j \circ f$ removed (ill-typed, [remark 3.4](#)); the compatibility square is now naturality of q , not a definition.
3. Composition of posteriors normalized; distributivity and $\text{Cond}/\otimes$ commutation corrected ([propositions 4.8](#) and [4.12](#)); chain rule stated with its conditional likelihood.
4. Observational equivalence: the congruence for $\otimes$ needs total compatibility, not q -monoidality; the congruence for Cond claimed earlier is false and is replaced by [proposition 5.5](#).
5. The comonad is now an instance of a general theorem ([theorem 6.1](#)); Sh_3 replaced by Sh ; naturality of δ is proved.
6. J is defined as $\text{Hyp}(j)$ so that CL1–CL3 are proved; the seventh tuple component of the earlier construction is no longer needed.
7. K/J coherence corrected to use D^3j (it was ill-typed with D^2j).
8. $\lambda^{[3]}$ had four components instead of five and an ill-typed shadow; it is replaced by λ^σ . The earlier unit and comultiplication theorems were false ([theorem 8.3](#)). The averaging law $\bar{\lambda}$ and [lemma 8.1](#) are new.
9. The weak distributive law and 2-cell α are withdrawn ([remark 8.9](#)). The contradiction between the claimed proof of hexagonal coherence and its listing as an open problem is resolved by removal.
10. Composite identifiability (which trivializes Hyp) removed; it is not needed. The “iff” proof corrected ([theorem 9.3](#)); condition (c) of nondegeneracy (finiteness) dropped as automatic.
11. Bibliography corrected. The earlier references “Böhm and Lack, *Weak distributive laws*, J. Algebra 282 (2004)” and “Lack, *Composing monads using semimonads*, J. Algebra 281 (2004)” could not be located and are removed; they are replaced by [21, 10, 11, 15, 16, 19, 9]. The de Finetti reference now cites its journal version [13].
12. Empirical claims toned down: no parity claim with TabPFN; BBP attribution removed; miscalibration flagged.

Items for the author to confirm.

13. Source and protocol of the TabPFN reference values in the benchmark table.
14. The nominal number of directions ($r = 3$ in the setup) versus the 6.6 planted directions in the spectral pre-pass.
15. Bootstrap standard errors for the chain-pooling curve.
16. Bibliographic details of the four self-citations [3, 4, 5, 6], which are reproduced as given in the previous version.

C Computational checks

All identities and counterexamples in [sections 5](#) and [8](#), [theorem 6.1](#) and [proposition 7.2](#) were checked in exact rational arithmetic on random instances with $X = \{0, 1\}$, two structural descriptors, and maps to a three-element set (one injective, one not). The checks confirm the comonad laws and naturality of ε, δ ; K/J coherence; naturality, (U), (Co) and (M) for $\bar{\lambda}$; the failure of (Cm) for $\bar{\lambda}$ on the explicit example of [theorem 8.7](#); naturality and (Co) for λ^σ and the failures of (U), (M), (Cm); and the counterexample of [proposition 5.5](#). The script `check_tower.py` accompanies this paper.

References

- [1] S. Abramsky and A. Brandenburger. The sheaf-theoretic structure of non-locality and contextuality. *New Journal of Physics*, 13:113036, 2011.
- [2] S. Abramsky, S. Mansfield, and R. S. Barbosa. The cohomology of non-locality and contextuality. *Electronic Proceedings in Theoretical Computer Science*, 95:1–14, 2012.
- [3] V. K. Awasthi. CausalIDView: gradient-free causal identification with structured shadows. Technical report, Department of Applied Mathematics and Statistics, Johns Hopkins University, 2026.
- [4] V. K. Awasthi. Recovering latent hierarchy without gradient descent. Technical report, Johns Hopkins University, September 2026. DOI: 10.13140/RG.2.2.24981.08167.
- [5] V. K. Awasthi. Holonomy obstructions for composed local operations. Technical report, Johns Hopkins University, September 2026. DOI: 10.13140/RG.2.2.25336.20483.
- [6] V. K. Awasthi. Three layers of realisation: hierarchy recovery, holonomy, information and contextuality. Technical report, Johns Hopkins University, September 2026. DOI: 10.13140/RG.2.2.24274.52163.
- [7] J. Baik, G. Ben Arous, and S. Péché. Phase transition of the largest eigenvalue for nonnull complex sample covariance matrices. *Annals of Probability*, 33(5):1643–1697, 2005.
- [8] P. Baudot and D. Bennequin. The homological nature of entropy. *Entropy*, 17(5):3253–3318, 2015.
- [9] J. Beck. Distributive laws. In *Seminar on Triples and Categorical Homology Theory*, volume 80 of *Lecture Notes in Mathematics*, pages 119–140. Springer, 1969.

- [10] G. Böhm. The weak theory of monads. *Advances in Mathematics*, 225(1):1–32, 2010. doi:10.1016/j.aim.2010.02.015.
- [11] G. Böhm, S. Lack, and R. Street. On the 2-categories of weak distributive laws. *Communications in Algebra*, 39(12):4567–4583, 2011. doi:10.1080/00927872.2011.616436.
- [12] T. Fritz. A synthetic approach to Markov kernels, conditional independence and theorems on sufficient statistics. *Advances in Mathematics*, 370:107239, 2020.
- [13] T. Fritz, T. Gonda, and P. Perrone. De Finetti’s theorem in categorical probability. *Journal of Stochastic Analysis*, 2(4):Article 6, 2021. doi:10.31390/josa.2.4.06. arXiv:2105.02639.
- [14] T. Fritz, T. Gonda, P. Perrone, and E. F. Rischel. Representable Markov categories and comparison of statistical experiments in categorical probability. *Theoretical Computer Science*, 961:113896, 2023.
- [15] R. Garner. The Vietoris monad and weak distributive laws. *Applied Categorical Structures*, 28(2):339–354, 2020. doi:10.1007/s10485-019-09582-w.
- [16] A. Goy and D. Petrişan. Combining probabilistic and non-deterministic choice via weak distributive laws. In *Proceedings of the 35th Annual ACM/IEEE Symposium on Logic in Computer Science (LICS)*, pages 454–464, 2020.
- [17] N. Hollmann, S. Müller, K. Eggersperger, and F. Hutter. TabPFN: A transformer that solves small tabular classification problems in a second. In *International Conference on Learning Representations (ICLR)*, 2023.
- [18] C. F. Manski. Nonparametric bounds on treatment effects. *American Economic Review*, 80(2):319–323, 1990.
- [19] J. Power and H. Watanabe. Combining a monad and a comonad. *Theoretical Computer Science*, 280(1–2):137–162, 2002. doi:10.1016/S0304-3975(01)00024-X.
- [20] R. Street. The formal theory of monads. *Journal of Pure and Applied Algebra*, 2(2):149–168, 1972.
- [21] R. Street. Weak distributive laws. *Theory and Applications of Categories*, 22(12):313–320, 2009.
- [22] J. P. Vigneaux. *Topology of Statistical Systems: A Cohomological Approach to Information Theory*. PhD thesis, Université Paris Diderot, 2019.